Rapid Sequence Intubation

CLINICAL INDICATIONS:

- Airway Compromise or Inability to Protect Airway
- Respiratory Failure (Hypoxic, Hypercapneic)
- Expected Clinical Deterioration
  - >40% TBSA Burns, Severe Sepsis, TBI with AMS, etc
- Patient or Crew Safety
  - Combative, prolong transfer in critically sick, etc

CONTRAINDICATIONS:

- High likelihood of failure (Distorted Anatomy)
- Penetrating neck trauma

PROCEDURE:

- Make a plan, prepare patient and equipment (See **PRE-INTUBATION CHECLIST**)

Conduct seven "P" pneumonic **(7Ps):**

PREPARE	☐ **SOAPME:** - Suction: available, check for function - Oxygen: Pre-Oxygenation + Apneic Oxygenation - Airways: ETT, SGA (iGel, King, etc), Cricothyrotomy - Pharmacology: Induction, Paralysis, Post-intubation Sedation - Monitor: BP, HR, RR, SpO2%, etCO2 capnography, 4-lead - Equipment: Bougie, Laryngoscope, Video Laryngoscope, Cric Kit ☐ **Difficult Airway Evaluation** (LEMON or HEAVEN Criteria) - Consider alternate airway, cricothyrotomy, or modify plan ☐ **Evaluate Cricothyrotomy Landmarks and Assess Procedural Difficulty**
PRE-OX	☐ **PreOxygenate / Denitrogenate** ≥ 3 minutes or 8 Vital Capacity Breaths with 15 LPM NRB or BVM + PEEP, and NC 4-6 LPM ☐ **Oxygenated ≥ 94%** - Patients remaining <94% may require CPAP or BVM + PEEP safe O2 saturation ☐ **Apneic Oxygenation with NC 15 LPM** once Induced/Sedated
POSITIONING	☐ **30° Head-up for Pre-Oxygenation** ☐ **Ear-to-Sternal Notch for Intubation** ☐ **C-Spine Consideration:** Open front of C-Collar; perform Manual In-line Stabilization
PRETREAT	☐ **Resuscitate** with IVF or Blood Products and Push-Dose Pressors to ensure SBP>100mmHg ☐ **3 - 5 Minute prior to Sedative / Paralytic** - **Fentanyl** 3mcg/kg IV to prevent Hypertension in head injury, cardiac ischemia, or aortic dissection - **Atropine** 0.02 mg/kg IV to prevent bradycardia in Peds (age <1y)
PARALYZE / SEDATE	☐ **Push** Ketamine or Etomidate; and then Rocuronium or Vecuronium ☐ **Apneic Oxygenate:** Turn Nasal Cannula to 15LPM once patient becomes drowsy ☐ **Monitor SpO2%** and Wait 45-60 second for adequate paralysis
PASS TUBE	☐ **Visualize Cords and Pass Tube**
POST-TUBE MANAGEMENT	☐ **Inflate Bulb and Begin Bagging** ☐ **Verify Tube Place with etCO2 waveform capnography** - Direct visualization, mist in tube, equal rise and fall, bilateral breath sounds with absent gastric sounds, improving SpO2%, etCO2 colorimetric gold color change x6 ☐ **Secure Endotracheal Tube with commercial securing device** ☐ **Place patient on Post-intubation Sedation**

RSI MEDICATIONS

Induction Agents:

Ketamine 1-2 mg/kg IV
Etomidate 0.3 mg/kg IV
Propofol 1-2.5 mg/kg IV

Paralytics:

Rocuronium 1.2 mg/kg IV
Vecuronium 0.1 mg/kg IV
Succinylcholine 1.5 mg/kg IV

Maintenance Sedation:

Ketamine 1-2mg/kg bolus, then 0.5-1mg/kg/hr
Propofol 10-75 mcg/kg/min
Midazolam 0.5 mg/kg bolus, then 0.01-0.1 mg/kg/hr

Push Dose Epi:

Epinehprine 5-20mcg IV q2-5min

Pre-Intubation Checklist

PRE-INTUBATION CHECKLIST	
INSTRUCTIONS FOR USE The Pre-Intubation Checklist serves as a final reference prior to administering a sedative and paralytic during a Rapid Sequence Intubation. Run the checklist to ensure preparation is complete before embarking on induction and paralysis. Refer to **RAPID SEQUENCE INTUBATION** for how to do an RSI.	
PLAN	☐ **Consider Dangerous Physiology Issue: SBP <100, SpO2 <94%, Metabolic Acidosis** ➢ Appropriately modify or avoid RSI in unstable patient ➢ **SBP < 100** → Consider resuscitation with IVF and vasopressors, Lower the dose of sedative ➢ **Sat < 94%** → Consider use of CPAP or BVM with PEEP valve to increase Mean Airway Pressure during Pre-Oxygenation ➢ **Severe Met. Acidosis** → Consider awake intubation or delayed sequence intubation with severe Metabolic Acidosis ☐ **Difficult Airway Evaluation** (LEMON, HEAVEN Criteria) ➢ Consider alternate airway, cricothyrotomy, or modify plan ☐ **Rapid Sequence Intubation, Delayed Sequence Intubation, Rapid Sequence Airway (SGA), or Cricothyrotomy** ➢ Choose the most appropriate technique for physiology and anatomy ☐ **Evaluate Cricothyrotomy Landmarks and Assess Procedural Difficulty** ☐ **Induction Agent / Paralytic** ➢ Choose and draw up appropriate Sedative and Paralytic ☐ **Push-Dose Pressors** ➢ Consider drawing up or administering Push-dose Epinephrine ☐ **Post-tube Sedation / Analgesia** ➢ Prepare Post-intubation Sedation and Analgesia ☐ **Consider Pretreatment 3-5 minutes prior** ➢ Fentanyl (TBI, CVA, MI, Ao Dissection); Atropine (Pediatric) ☐ **Failed Airway Plan Verbalized to the team** ➢ Discuss management plan for failed intubation
PATIENT PREPARATION	☐ **PreOx** ≥ 3 minutes with 15 LPM NRB or BVM + PEEP, and NC 4-6 LPM ☐ **Apneic Oxygenation with NC 15 LPM** once Induced/Sedated ☐ **Oxygenated ≥ 94%** prior to Induction ➢ Consider using CPAP or BVM + PEEP if unable to reach 94% with NRB ☐ **Positioning**: 30° Head-up for Pre-Ox, Ear-to-Sternal Notch for Intubation ➢ If C-Spine Consideration, open front of C-Collar and perform Manual In-line Stabilization ☐ **Monitor is Visible** (HR, BP, SpO2%, RR, etCO2) ☐ **Reliable IV Access Tested**
EQUIPMENT	☐ BVM (± PEEP Valve) on Oxygen ☐ Waveform Capnography on BVM (minimum Colorimetric) ☐ ± Video Laryngoscope powered on ☐ ± Back-up Laryngoscope ETT, ETT size down, 2x Stylet, 2x Syringe, Tube Securing ☐ Bougie ☐ OPA, NPA, SGA (iGel, LMA, King LT) ☐ Nasogastric or Orogastric Tube ☐ Cricothyrotomy Kit ☐ Suction on and accessible